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United States Patent	12402421
Kind Code	B2
Date of Patent	August 26, 2025
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Surface uniformity control in pixel structures of image sensors

Abstract

A semiconductor device with an image sensor and a method of fabricating the same are disclosed. The method includes depositing a dielectric layer on a substrate, forming a trench within the dielectric layer and the substrate, forming an epitaxial structure within the trench, and forming a barrier layer with first and second layer portions. The first layer portion is formed on a sidewall portion of the trench that is not covered by the epitaxial structure. The method further includes forming a capping layer on the epitaxial structure and adjacent to the barrier layer, selectively doping regions of the epitaxial structure and the capping layer, selectively forming a silicide layer on the doped regions, depositing an etch stop layer on the silicide layer, and forming conductive plugs on the silicide layer through the etch stop layer.

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Appl. No.: 18/231471

Filed: August 08, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230395643 A1	Dec. 07, 2023

Related U.S. Application Data

division parent-doc US 17102623 20201124 PENDING child-doc US 18231471
us-provisional-application US 63031752 20200529

Publication Classification

Int. Cl.: H10F39/12 (20250101); H10F39/00 (20250101); H10F71/00 (20250101); H10F77/122 (20250101); H10F77/30 (20250101)

U.S. Cl.:

CPC H10F39/011 (20250101); H10F39/024 (20250101); H10F39/199 (20250101); H10F39/8027 (20250101); H10F39/805 (20250101); H10F71/121 (20250101); H10F77/122 (20250101); H10F77/306 (20250101);

Field of Classification Search

CPC: H01L (27/14607); H01L (27/1462); H01L (31/028); H01L (31/02161); H01L (31/1808-1816); H10F (39/8027); H10F (39/805)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/102,623, titled “Surface Uniformity Control in Pixel Structures of Image Sensors,” filed Nov. 24, 2020, which claims the benefit of U.S. Provisional Patent Application No. 63/031,752 titled “A Flat Device Structure for IR Sensors,” filed May 29, 2020, each of which is incorporated by reference herein in its entirety.

BACKGROUND

(1) Semiconductor image sensors are used to sense incoming visible or non-visible radiation, such as visible light and infrared light. Complementary metal-oxide-semiconductor (CMOS) image sensors (CIS) and charge-coupled device (CCD) sensors are used in various applications, such as digital still cameras, mobile phones, tablets, and goggles. These image sensors utilize an array of pixels that absorb (e.g., sense) the incoming radiation and convert it into electrical signals. An example of an image sensor is a backside illuminated (BSI) image sensor, which detects radiation from a “backside” of a substrate of the BSI image sensor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of this disclosure are best understood from the following detailed description when read with the accompanying figures.
- (2) FIG. 1A illustrates a cross-sectional view of a semiconductor device with a BSI image sensor, in accordance with some embodiments.
- (3) FIGS. 1B-1F illustrate cross-sectional views of pixel structures of a BSI image sensor, in accordance with some embodiments.
- (4) FIG. 2 is a flow diagram of a method for fabricating a semiconductor device with a BSI image sensor, in accordance with some embodiments.
- (5) FIGS. 3-37 illustrate cross-sectional views of a semiconductor device with a BSI image sensor at various stages of its fabrication process, in accordance with some embodiments.
- (6) Illustrative embodiments will now be described with reference to the accompanying drawings. In the drawings, like reference numerals generally indicate identical, functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

(7) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the process for forming a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. As used herein, the formation of a first feature on a second feature means the first feature is formed in direct contact with the second feature. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(8) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(9) It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “exemplary,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

(10) It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.

(11) In some embodiments, the terms “about” and “substantially” can indicate a value of a given quantity that varies within 5% of the value (e.g., $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$ of the value). These values are merely examples and are not intended to be limiting. The terms “about” and “substantially” can refer to a percentage of the values as interpreted by those skilled in relevant art(s) in light of the teachings herein.

(12) A BSI image sensor (e.g., time-of-flight sensor) includes a pixel region (also referred to as radiation-sensing region”) with an array of pixel structures formed on a substrate (e.g., a semiconductor substrate). The pixel structures are configured to receive a radiation (e.g., infra-red radiation) reflected from an object and convert photons from the received radiation to electrical signal. The electrical signal is used to produce a depth image by measuring the phase-delay of the received radiation. The electrical signal is subsequently distributed to processing components attached to the BSI image sensor. For this reason, the pixel structures overlie a multi-level metallization layer configured to distribute the electrical signal generated within the pixel structures to appropriate processing components.

(13) The multi-level metallization layer is coupled to a first surface of the substrate, which is also referred to as the “front side” surface of the substrate. The pixel structures are formed on the front side surface of the substrate and the radiation is received by the pixel structures through a second surface of the substrate that is opposite to the front side surface of the substrate. This second surface of the substrate is also referred to as the “back side” surface of the substrate. Each of the pixel structures includes a germanium (Ge) or silicon germanium (SiGe) epitaxial structure disposed within the substrate, a silicon capping layer disposed on the Ge or SiGe epitaxial structure, and doped regions disposed within the Ge or SiGe epitaxial structure and the silicon capping layer. The silicon capping layers passivate the Ge or SiGe epitaxial structures and provide silicon atoms for the formation of silicide structures on the doped regions.

(14) The BSI image sensor can be electrically coupled to external devices (e.g., an external circuitry) through wire connectors attached to pad structures formed on the back side surface of the substrate. To achieve this, the pad structures of the BSI image sensor extend from the back side surface of the substrate to the front side surface of the substrate and electrically connect to the multi-level metallization layer. Accordingly, the multilevel metallization layer, which provides electrical signal connection to the BSI image sensor can be electrically connected to an external device or circuit through the pad structures. The pad structures can be disposed at the periphery of the BSI image sensor around the pixel region.

(15) A challenge with BSI image sensors is reducing or eliminating dark current, which is induced by surface non-uniformity of the Ge or SiGe epitaxial structures formed on the front side surface of the substrate. The dark current is an electrical current that passes through the Ge or SiGe epitaxial structures even when no photons are received by the BSI image sensors. The dark current generation source is the non-uniform interfaces between the Ge or SiGe epitaxial structures and the silicon capping layers that induce charge carriers (e.g., holes) to accumulate at the non-uniform interfaces. Such dark current causes the Ge or SiGe epitaxial structures to generate electrical signals that do not correspond to the actual amount of the radiation received by the BSI image sensors. As a result, the dark current degrades the performance of the BSI image sensors in accurately determining the distance of the object from the BSI image sensor.

(16) The surface non-uniformity, such as recesses, are formed near the edges of the Ge or SiGe epitaxial structures as a result of surface deformation in the Ge or SiGe epitaxial structures caused by the high temperature (e.g., temperature greater than 500° C.) processing of overlying layers, such as the silicon capping layers. The high temperature can cause the Ge or SiGe materials near the edges of the Ge or SiGe epitaxial structures to become ductile and laterally flow to form Ge or SiGe layers over the edges of the trenches in which the Ge or SiGe epitaxial structures are formed. These trench edges can be substrate edges on the front side surface of the substrate. The non-uniform surfaces can also lead to the formation of gaps in the overlying layers formed on the Ge or SiGe epitaxial structures. These gaps can introduce processing chemicals (e.g., etching solutions) into the Ge or SiGe epitaxial structures during the processing of the overlying layers and damage the Ge or SiGe epitaxial structures.

(17) The present disclosure provides example methods of fabricating BSI image sensors with improved surface uniformity of Ge or SiGe epitaxial structures in pixel structures. In some embodiments, the example methods form barrier layers at the edges of the trenches in which the Ge or SiGe epitaxial structures are formed. The barrier layers can prevent the formation of Ge or SiGe layers at the trench edges because the barrier layers include dielectric materials, such as silicon oxide (SiO₂), that are not favorable for the growth of Ge or SiGe layers. The materials of the barrier layers can inhibit the growth of Ge or SiGe layers on the barrier layers, and as a result, inhibits the growth of Ge or SiGe layers at the trench edges. Preventing the lateral expansion of the materials of the Ge or SiGe epitaxial structures can prevent the formation of recesses on the surfaces of the Ge or SiGe epitaxial structures.

(18) Thus, the barrier layers at the trench edges can minimize or eliminate surface non-uniformity, and consequently reduce or eliminate the formation of non-uniform interfaces between the Ge or SiGe epitaxial structures and silicon capping layers. In some embodiments, the surface non-uniformity in the Ge or SiGe epitaxial structures is reduced by about 50% to about 100% compared to epitaxial structures in BSI image sensors without the barrier layers. As a result, the sensor performance of the BSI image sensors with the barrier layers is increased by about 40% to about 60% compared to BSI image sensors without the barrier layers at the trench edges.

(19) A semiconductor device **100** having a BSI image sensor **102**, a first a multi-level metallization layer **104**, a second a multi-level metallization layer **105**, and an application specific integrated circuit (ASIC) **106** is described with reference to FIGS. **1A-1F**, according to some embodiments. FIG. **1A** illustrates a cross-sectional view of semiconductor device **100**, according to some embodiments. FIGS. **1B-1F** illustrate enlarged cross-sectional views of region **102A1** of BSI image sensor **102** in FIG. **1A**, according to various embodiments. The discussion of elements in FIGS. **1A-1D** with the same annotations applies to each other, unless mentioned otherwise.

(20) Referring to FIG. **1A**, BSI image sensor **102** can be disposed on and electrically coupled to first multi-level metallization layer **104**. First multi-level metallization layer **104** can be disposed on second a multi-level metallization layer **105**, which can be disposed on and electrically coupled to ASIC **106**. First multi-level metallization layer **104** can include a multi-level interconnect structure **104A** embedded in an inter-metal dielectric (IMD) layer **104B**, which is disposed on a

bonding layer **104C** with metal lines **104D**. Similarly, second multi-level metallization layer **105** can include a multi-level interconnect structure **105A** embedded in an IMD layer **105B** and a bonding layer **105C** with metal lines **105D** disposed on IMD layer **105B**. Bonding layers **104C-105C** can be bonded to each other by a suitable bonding method, such as direct bonding, eutectic bonding, hybrid bonding, and optical fusion bonding, and can be electrically connected to each other through metal lines **104D-105D**. As a result, BSI image sensor **102** can be electrically connected to ASIC **106** through first and second multi-level metallization layers **104-105**. ASIC **106** can include active devices **106A** (e.g., transistor structures) to form logic and memory circuits. In some embodiments, active devices **106A** can be configured to process electrical signals received from BSI image sensor **102**.

(21) BSI image sensor **102** can be formed on a substrate **108** with a front side surface **108a** and a back side surface **108b**. In some embodiments, substrate **108** can include a monocrystalline silicon substrate. In some embodiments, substrate **108** can include a semiconductor material, such as Si, Ge, SiGe, silicon carbide (SiC), indium phosphide (InP), gallium arsenide (GaAs), silicon arsenide (SiAs), gallium phosphide (GaP), indium phosphide (InP), silicon germanium carbide (SiGeC), germanium stannum (GeSn), silicon germanium stannum (SiGeSn), gallium arsenic phosphide (GaAsP), gallium indium phosphide (GaInP), gallium indium arsenide (GaInAs), gallium indium arsenic phosphide (GaInAsP), and any other suitable semiconductor material. In some embodiments, substrate **108** can include a silicon-on-insulator (SOI) structure or a germanium-on-insulator (GOI) structure. Other suitable materials for substrate **108** are within the scope of the present disclosure.

(22) Referring to FIG. 1A, BSI image sensor **102** can include a stack of layers **110** disposed between front side surface **108a** and first multi-level metallization layer **104**. In some embodiments, stack of layers **110** can include (i) a passivation layer **112** disposed on multi-level metallization layer **104**, (ii) an interlayer dielectric (ILD) layer **114** disposed on passivation layer **112**, (iii) an etch stop layer (ESL) **116** disposed on ILD layer **114**, and (iv) a dielectric layer **118** disposed on ESL **116**. BSI image sensor **102** can further include a pixel region **102A**, isolation regions **102B**, and a contact pad region **102C**, according to some embodiments.

(23) In some embodiments, pixel region **102A** can include an array of pixel structures **122A-122B**. Though an array of two pixel structures **122A-122B** are shown, BSI image sensor **102** can have any number of pixel structures arranged in a one-dimensional array or a two-dimensional array. Pixel structures **122A-122B** are configured to receive incident radiation beams **123** through microlens **125** on back side surface **108b** and convert radiation beams **123** to an electrical signal. The electrical signal is distributed by pad structure **120** and first and second multi-level metallization layers **104-105** to ASIC **106** and/or other external circuits.

(24) In some embodiments, BSI image sensor **102** can be a time-of-flight sensor configured to determine the distance of an object from BSI image sensor **102** based on the known speed of light. For example, a light pulse generator (not shown) disposed on or near the BSI image sensor can project a light pulse (e.g., near infra-red radiation) on the object and the light pulse reflected by the object can be detected by pixel structures **122A-122B**. Based on the time difference between the projection time of the light pulse and the detection time of the reflected light pulse, the distance of the object from BSI image sensor **102** can be determined.

(25) Pixel structures **122A-122B** are electrically isolated from each other by dielectric layer **118** and are protected by passivation layer **112**, ILD layer **114**, and ESL **116** during fabrication of BSI image sensor **102**. In some embodiments, dielectric layer **118** can include a nitride layer, an oxide layer, an oxynitride layer, or a suitable dielectric material. In some embodiments, dielectric layer **118** can include an oxide of a material of substrate **108**, such as silicon oxide (SiO_{sub}.x). In some embodiments, ESL **116** can include a nitride layer, an oxide layer, an oxynitride layer, a carbide layer, or a suitable dielectric material. In some embodiments, ESL **116** can include silicon nitride (SiN_x), silicon oxide (SiO_x), silicon oxynitride (SiON), silicon carbide (SiC), silicon carbonitride

(SiCN), boron nitride (BN), silicon boron nitride (SiBN), silicon carbon boron nitride (SiCBN), or a combination thereof. In some embodiments, ILD layer **114** can include a low-k dielectric layer (e.g., a dielectric with a dielectric constant less than about 3.9), an ultra-low-k dielectric layer (e.g., a dielectric with a dielectric constant less than about 2.5), or an oxide layer (e.g., silicon oxide (SiO₂)). In some embodiments, passivation layer **112** can include a nitride layer, an oxide layer, an oxynitride layer, a polymer layer (e.g., polyimide or polybenzoxazole), or a combination thereof.

(26) In some embodiments, pixel structures **122A-122B** are similar to each other in structure and composition. The discussion of pixel structure **122A** applies to pixel **122B**, unless mentioned otherwise. Pixel structure **122A** is described with reference to FIGS. **1A-1F**, according to some embodiments. Pixel structure **122A** can have different cross-sectional views as illustrated in FIGS. **1B-1F**, according to various embodiments. FIGS. **1B** and **1D-1E** illustrate enlarged cross-sectional views of pixel structure **122A** within region **102A1** of FIG. **1A**, according to various embodiments. FIG. **1C** illustrates a cross-sectional view along line A-A and XY-plane of FIG. **1B** and FIG. **1F** illustrates a cross-sectional view along line B-B and XY-plane of FIGS. **1D-1E**. The cross-sectional views of FIGS. **1C** and **1F** are referred to as “front side views” of pixel structure **122A**.

(27) Referring to FIGS. **1A-1F**, in some embodiments, pixel structure **122A** can include (i) an epitaxial structure **124** disposed within substrate **108**, (ii) a capping layer **126** disposed on epitaxial structure **124**, (iii) a barrier layer **128** disposed on the edges (not visible in FIGS. **1A-1C**; discussed in further details below) of substrate **108** adjacent to and surrounding epitaxial structure **124** and capping layer **126**, (iv) p-type doped regions **130** disposed within epitaxial structure **124** and capping layer **126**, (v) n-type doped region **132** disposed within epitaxial structure **124** and capping layer **126**, (vi) contact structures **134** disposed on p- and n-type doped regions **130-132**, and (vii) vias **136** disposed on contact structures **134**.

(28) Epitaxial structure **124** can be formed on front side surface **108a** and can include a group IV element (e.g., Si, Ge, etc.) of the periodic table. In some embodiments, epitaxial structure **124** can include a group IV element that is different from a group IV element of substrate **108**. In some embodiments, epitaxial structures **124** can include undoped Ge or SiGe. In some embodiments, front side surface **124f** of epitaxial structure **124** can be spaced apart from front side surface **108a** of substrate **108** by a distance **D1** along a Z-axis. Distance **D1** can range from about 50 nm to about 100 nm or can be other suitable dimensions.

(29) Capping layer **126** can include a group IV element (e.g., Si) of the periodic table that is different from the element included in epitaxial structure **124**. The element included in capping layer **126** can have a band gap different from the band gap of the element included in epitaxial structure **124**, which results in band discontinuity between epitaxial structure **124** and capping layer **126** (e.g., a difference between the minimum conduction band energy and/or the maximum valence band energy of epitaxial structure **124** and capping layer **126**). In some embodiments, front side surface **126f** of capping layer **126** can be substantially coplanar with front side surface **108a** of substrate **108**, as shown in FIGS. **1B** and **1D**, can be at an XY-plane higher than that of front side surface **108a** (not shown), or can be at an XY-plane lower than that of front side surface **108a**, as shown in FIG. **1E**. In some embodiments, capping layer **126** can extend over barrier layer **128** by a distance **D2** along a Z-axis, as shown in FIG. **1E**. Distance **D2** can range from about 5 nm to about 10 nm or can be other suitable dimensions.

(30) The profiles of sidewalls **126s** of capping layer **126** can depend on the method of forming capping layer **126**. In some embodiments, sidewalls **126s** can have profiles as shown in FIG. **1B** when capping layer **126** is grown on epitaxial structure **124** by an epitaxial process. Sidewalls **126s** of epitaxially grown capping layer **126** and sidewalls **128s** of barrier layer **128** are in physical contact at and near front side surface **124f**. But as sidewalls **126s** and **128s** extend towards front side surface **126f**, sidewalls **126s** and **128s** can become spaced apart from each other by a gap **127** (also referred to as dielectric material-filled gap **127**), which is filled with a portion of dielectric

layer **118**, as shown in FIGS. **1B-1C**. Gap **127** can be formed because the material of barrier layer **128** inhibits the epitaxial growth of capping layer **126** on barrier layer **128**. In some embodiments, sidewalls **126s** can have profiles as shown in FIGS. **1D-1E** when capping layer **126** is formed on epitaxial structure **124** by a deposition process. The profiles of sidewalls **126s** of deposited capping layer **126** can follow the profiles of sidewalls **128s**, as shown in FIGS. **1D-1E**.

(31) Barrier layer **128** can include an oxide of the material of substrate **108** (e.g., SiO₂) or a suitable dielectric material. In some embodiments, barrier layer **128** can be formed at the edges of front side surface **108a** surrounding capping layer **126**. A portion of barrier layer **128** can extend along a portion of front side surface **108a** and along a portion of sidewall **108s** of substrate **108**. Surfaces **128b** of barrier layer **128** can be partially disposed on top edges of epitaxial structure **124**, as shown in FIGS. **1B** and **1D-1E**, when barrier layer **128** is grown on substrate **108** by an oxidation process. On the other hand, surfaces **128b** of barrier layer **128** can be fully disposed on top edges of epitaxial structure **124**, as shown in FIGS. **29** and **33**, when barrier layer **128** is formed by a deposition process. In some embodiments, barrier layer **128** can have a thickness along an X-axis ranging from about 0.5 nm to about 5 nm. Thickness of barrier layer **128** below 0.5 nm may not adequately prevent the deformation of epitaxial structure **124** during subsequent high temperature processes. On the other hand, if thickness is above 5 nm, the processing time (e.g., deposition time or oxidation time) of barrier layer increases, and consequently increases device manufacturing cost.

(32) Contact structures **134** can be configured to electrically connect epitaxial structure **124** to first multi-level metallization layer **104** through via structures **136**. Each of contact structures **134** can include a silicide layer **134A** and a contact plug **134B**. Silicide layers **134A** are disposed on p- and n-type doped regions **130-132** and within capping layer **126**. Surfaces **134As** of silicide layers **134A** can be substantially coplanar (not shown) with front side surface **126f** of capping layer **126** or can extend below front side surface **126f**, as shown in FIGS. **1A-1B** and **1D-1E**. In some embodiments, silicide layers **134A** can include nickel silicide (NiSi), tungsten silicide (WSi₂), titanium silicide (TiSi₂), cobalt silicide (CoSi₂), or a suitable metal silicide. Contact plugs **134B** are disposed on silicide layers **134A** and within ILD layer **114**. In some embodiments, contact plugs **134B** can include conductive materials, such as ruthenium (Ru), iridium (Ir), nickel (Ni), Osmium (Os), rhodium (Rh), aluminum (Al), molybdenum (Mo), tungsten (W), cobalt (Co), copper (Cu), and any suitable metallic material. Via structures **136** are disposed on contact plugs **134B** and within passivation layer **112**. In some embodiments, via structures **136** can include conductive materials, such as Ru, Co, Ni, Al, Mo, W, Ir, Os, Cu, Pt, and any other suitable metallic material.

(33) Referring to FIG. **1A**, isolation regions **102B** can include n-type doped regions **140** and p-type doped regions **141** that are configured to form PN junction based isolation structures. The isolation structures can be electrically connected to first multi-level metallization layer **104** and/or other circuits through contact structures **134** and via structures **136**. Contact pad region **102C** can include a pad structure **120** and one or more conductive bonding pads or solder bumps (not shown) on pad structure **120** through which electrical connections between BSI image sensor **102** and external circuit can be established. Pad structure **120** is an input/output (I/O) port of BSI image sensor **102** and includes a conductive layer that is electrically coupled to multi-level interconnect structure **104A**.

(34) FIG. **2** is a flow diagram of an example method **200** for fabricating semiconductor device **100**, according to some embodiments. For illustrative purposes, the operations illustrated in FIG. **2** will be described with reference to the example fabrication process for semiconductor device **100** as illustrated in FIGS. **3-33**. FIGS. **3-33** are cross-sectional views of semiconductor device **100** at various stages of fabrication, according to some embodiments. Operations can be performed in a different order or not performed depending on specific applications. It should be noted that method **200** may not produce a complete semiconductor device **100**. Accordingly, it is understood that

additional processes can be provided before, during, and after method **200**, and that some other processes may only be briefly described herein. Elements in FIGS. **3-33** with the same annotations as elements in FIGS. **1A-1F** are described above.

(35) In operation **205**, isolation structures are formed on a front side surface of a substrate. For example, as shown in FIG. **3**, n-type doped regions **140A** and p-type doped regions **140B** of isolation structures **140** are formed on front side surface **108a** of substrate **108**. N- and p-type doped regions **140A-140B** can be formed by ion implanting dopants into substrate **108** through front side surface **108a**. Following the formation of doped regions **140A-140B**, a dielectric layer **318** can be deposited on front side surface **108a**, as shown in FIG. **3**. Dielectric layer **318** can include a material of dielectric layer **118**. The formation of dielectric layer **318** can include using a CVD process, an ALD process, a thermal oxidation process, or a suitable deposition process for dielectric materials.

(36) Referring to FIG. **2**, in operation **210**, epitaxial structures are formed on the front side surface of the substrate. For example, as described with reference to FIGS. **4-9**, epitaxial structures **124** can be formed at the same time on front side surface **108a** of substrate **108**. The formation of epitaxial structures **124** can include sequential operations of (i) forming trenches **424** at the same time in substrate **108** through dielectric layer **318**, as shown in FIG. **4**, (ii) epitaxially growing, at the same time, structures **524** within trenches **424** of FIG. **4**, as shown in FIG. **5**, (iii) performing a CMP process on the structure of FIG. **5**, as shown in FIG. **6**, to form epitaxial structures **724** within trenches **424** of FIG. **4**, as shown in FIG. **7**, (iv) etching epitaxial structures **724** of FIG. **7** to form epitaxial structures **124** with heights H1-H2, as shown in FIG. **8**, and (v) performing a cleaning process on the structure of FIG. **8**, which can result in the structure shown in FIG. **9**.

(37) The formation of trenches **424** can include a dry etching process with etchants, such as chlorine-based gas, helium, fluorine-based gas, argon, and a combination thereof. The epitaxial growth of structures **524** can include epitaxially growing monocrystalline or polycrystalline structures of a semiconductor material, such as a group IV element (e.g., Si, Ge, etc.) of the periodic table. The CMP process can include using a CMP slurry with a higher removal selectivity for the material of structures **524** than for the material of dielectric layer **318**. The term “removal selectivity” refers to the ratio of the removal rates of two different materials under the same removal conditions. In some embodiments, the CMP slurry can have a removal selectivity that is about 20 times to about 200 times greater for the material of structures **524** than for the material of dielectric layer **318**. The CMP slurry can include hydrogen peroxide, potassium peroxydisulfate, nitrogen-oxide-based compound, polyethylene glycol, abrasive particles, such as colloidal silica, fumed silica, aluminum oxide, and a combination thereof.

(38) In some embodiments, the CMP process can form a substantially flat surface profile (solid line) of front side surface **724f** of epitaxial structure **724** or a curved surface profile (dotted line) of front side surface **724f**, as shown in FIG. **7**. In some embodiments, front side surface **724f** can be substantially coplanar (not shown) or non-coplanar (shown in FIG. **7**) with front side surface of dielectric layer **318**.

(39) The etching of epitaxial structures **724** can include a wet etch process, a dry etch process, or a vapor etch process using halogen-based etchants. The etchants have a higher etch selectivity (e.g., about 20 to about 50 times higher) for the material of epitaxial structures **724** than the etch selectivity for the material of dielectric layer **318** and substrate **108**. The etching of epitaxial structures **724** to form epitaxial structures **124** can expose a sidewall portion **424s** of trench **424** (FIG. **4**) by lowering front side surface **124f** by a distance D1 along a Z-axis from front side surface **108a** of substrate **108**. Distance D1 can range from about 50 nm to about 100 nm or can be any other suitable dimensions. Heights H1-H2 of epitaxial structures **124** can be equal to or different from each other.

(40) The cleaning process can include cleaning the structure of FIG. **8** to remove contaminants and/or residues from the CMP process and/or the etching process with an acid-based cleaning

solution, such as diluted hydrofluoric acid (DHF), for a time period of about 20 seconds to about 60 seconds. In some embodiments, the DHF solution can have a water to HF ratio of about 50:1 to about 100:1. In some embodiments, the cleaning solution can etch portions of dielectric layer **318** to expose a surface portion **108s** of front side surface **108s** and can etch exposed portions of substrate **108** to round substrate edges (also referred to as trench edges **108e**), as shown in FIG. **9**. Thus, the etching and cleaning processes form exposed substrate regions **108r** that includes the exposed sidewall portions **424s**, surface portions **108s**, and rounded substrate edges **108e**, as shown in FIG. **9**.

(41) Referring to FIG. **2**, in operation **215**, barrier layers are formed at substrate edges that are adjacent to the epitaxial structures. For example, as shown in FIG. **10**, barrier layers **128** are formed on exposed substrate regions **108r** of FIG. **10**. Forming barrier layers **128** on exposed substrate regions **108r** can prevent the materials of epitaxial structures **124** from laterally expanding and reacting with the material of substrate regions **108r** during subsequent high temperature processes. The lateral expansion of epitaxial structures **124** can deform the surface profile of epitaxial structures **124**.

(42) The formation of barrier layers **128** can include performing an oxidation process on the structure of FIG. **9**. The oxidation process can include oxidizing exposed substrate regions **108r** of FIG. **9** with an oxidizing solution for a time period of about 60 seconds to about 120 seconds. The oxidizing solution can include deionized water and ozone or hydrogen peroxide (H.sub.2O.sub.2). In some embodiments, the oxidizing solution of deionized water and ozone can include about 10 parts per million (ppm) to about 100 ppm of ozone. In some embodiments, the oxidizing solution of H.sub.2O.sub.2 can include a water to H.sub.2O.sub.2 ratio of about 50:1 to about 100:1. In some embodiments, the oxidizing solution can oxidize a top portion of epitaxial structures **124** to form oxide layers **1024** on epitaxial structures **124**, as shown in FIG. **10**. Thus, oxide layers **1024** include an oxide of the material of epitaxial structures **124**.

(43) Following the formation of barrier layers **128**, an annealing process can be performed on the structure of FIG. **10** at a temperature of about 500° C. to about 800° C. in a hydrogen ambient. The annealing process can reduce oxide layers **1024** to the material of epitaxial structures **124**, as shown in FIG. **11**. For example, when epitaxial structures **124** include Ge and oxide layer **1024** includes germanium oxide (GeO.sub.x), the annealing process can reduce GeO.sub.x to Ge.

(44) Referring to FIG. **2**, in operation **220**, capping layers are formed on the epitaxial structures. For example, as shown in FIG. **12**, capping layers **126** can be formed on epitaxial structures **124**. In some embodiments, the formation of capping layers **126** can include epitaxially growing a Si, Ge, or SiGe layer on epitaxial structures **124**. The profiles and dimensions of capping layers **126** of FIG. **12** can be similar to that described above with reference to FIGS. **1B-1C**.

(45) In some embodiments, instead of epitaxially growing capping layers **126**, the formation of capping layers **126** can include sequential operations of (i) depositing a Si, Ge, or SiGe layer **1326** on the structure of FIG. **11** to form the structure of FIG. **13**, (ii) forming a patterned masking layer (not shown) on layer **1326** to protect portions of layer **1326** on epitaxial structures **124**, and (iii) selectively etching portions of layer **1326** that are not protected by the patterned masking layer to form the structure of FIG. **14**. The deposition of Si, Ge, or SiGe layer can include using a silicon precursor (e.g., silane (SiH.sub.4) or dichlorosilane (DCS)) and/or a germanium precursor (e.g., germane (GeH.sub.4)) in a chemical vapor deposition (CVD) process or an atomic layer deposition (ALD) process. The profiles and dimensions of capping layers **126** of FIG. **14** can be similar to that described above with reference to FIGS. **1D** and **1F**.

(46) In some embodiments, following the formation of capping layers **126**, a dielectric layer **818** with material similar to the material of dielectric layer **318** can be deposited on the structure of FIG. **12** to form the structure of FIG. **15**. The formation of dielectric layer **818** can include using a CVD process, an ALD process, or a thermal oxidation process.

(47) Referring to FIG. **2**, in operation **225**, doped regions are formed within the epitaxial structures

and the capping layers. For example, as described with reference to FIGS. **16-18**, p-type doped regions **130** and n-type doped regions **132** can be formed within capping layers **126** and epitaxial structures **124**. The formation of p-type doped regions **130** and n-type doped regions **132** can include sequential operations of (i) forming a patterned masking layer **1642** with openings **1644** on the structure of FIG. **15**, as shown in FIG. **16**, (ii) ion implanting n-type dopants **1646** into capping layers **126** and epitaxial structures **124** through openings **1644** to form doped regions **132**, as shown in FIG. **16**, (iii) removing patterned layer **1642** (not shown), (iv) forming a patterned masking layer **1748** with openings **1750**, as shown in FIG. **17**, (v) ion implanting p-type dopants **1752** into capping layers **126** and epitaxial structures **124** through openings **1750** to form p-type doped regions **130**, as shown in FIG. **17**, and (vii) performing an annealing process on the structure of FIG. **18** after removing patterned layer **1748** to activate the dopants in doped regions **130-132**.

(48) Referring to FIG. **2**, in operation **230**, contact structures and via structures are formed on the doped regions and the isolation structures. For example, as described with reference to FIGS. **19-23**, contact structures **134** with silicide layers **134A** and contact plugs **134B** are formed on doped regions **130-132** and **140A-140B**, and via structures **136** are formed on contact structures **134**. The formation of contact structures **134** can include sequential operations of (i) forming silicide openings **1954** on doped regions **130-132** and **140A-140B**, as shown in FIG. **19**, (ii) forming silicide layers **134A** on doped regions **130-132** and **140A-140B**, as shown FIG. **20**, (iii) depositing ESL **116** on the structure of FIG. **20** to form the structure of FIG. **21**, (iv) depositing ILD layer **114** on the structure of FIG. **21**, as shown in FIG. **22**, (v) forming contact openings **2256** within ILD layer **114** and ESL **116** to expose portions of silicide layers **134A**, as shown in FIG. **22**, and (vi), forming contact plugs **134B** within contact openings **2256**, as shown in FIG. **23**.

(49) The formation of silicide layers **134A** can include sequential operations of (i) depositing a metal layer (not shown) on the structure of FIG. **19**, (ii) performing an annealing process on the structure with the metal layer, and (iii) removing the non-reacted portions of the metal layer on dielectric layer **118** to form the structure of FIG. **20**. The formation of via structures **136** can include depositing a metal layer (not shown) on the structure after the formation of contact plugs **134B** and patterning the deposited metal layer to form the structure of FIG. **23**. Following the formation of via structures **136**, passivation layer **112** can be deposited on the structure of FIG. **23**, as shown in FIG. **24**.

(50) Referring to FIG. **2**, in operation **235**, a multi-level metallization layer is formed on the via structures and bonded to an integrated circuit. For example, as shown in FIG. **25**, first multi-level metallization layer **104** with multi-level interconnect structure **104A** embedded within IMD layer **104B** is formed on via structures **136** and passivation layer **112**. The formation of multi-level metallization layer **104** can be followed by bonding second multi-level metallization layer **105** and ASIC **106** to multi-level metallization layer **104**, as shown in FIG. **25**.

(51) Referring to FIG. **2**, in operation **240**, a pad structure is formed on the multi-level metallization layer through a back side surface of the substrate. For example, as shown in FIG. **26**, pad structure **120** is formed on multi-level interconnect structure **104A** through back side surface **108b**. The formation of pad structure **120** can include sequential operations of (i) forming a pad opening (not shown) within substrate **108**, dielectric layer **118**, ESL **116**, ILD layer **114**, passivation layer **112**, and a portion of IMD layer **104B**, (ii) depositing a conductive layer (not shown) within the pad opening, and (iii) patterning and etching the conductive layer to form pad structure **120** within pad opening, as shown in FIG. **26**. Following the formation of pad structure **120**, an array of microlens **125** can be formed on back side surface **108b**.

(52) In some embodiments, in operation **215** of FIG. **2**, instead of the oxidation process described to form barrier layers **128** with reference to FIG. **10**, barrier layers **128** can be formed as described with reference to FIGS. **27-29**. The formation of barrier layers **128** can include the sequential operations of (i) depositing an oxide layer **2728** on the structure of FIG. **9**, as shown in FIG. **27**, (ii) forming a patterned masking layer **2758** with openings **2760** on oxide layer **2728**, as shown in FIG.

27, (iii) etching the portions of oxide layer **2728** exposed through openings **2760** to form etched oxide layer **2828**, as shown in FIG. **28**, (iv) removing patterned masking layer **2758** to form the structure of FIG. **28**, and (v) etching the portions of oxide layer **2828** on dielectric layer **318** to form the structure of FIG. **29** on which operations **220-240** can be performed. In some embodiments, oxide layer **2728** can include an insulating material and can be deposited with a CVD process, an ALD process, or a suitable deposition process for depositing dielectric materials. In some embodiments, the etching of oxide layers **2728** and **2828** can be performed using a wet etch process (e.g., using DHF) or a dry etch process. Following the formation of barrier layers **128**, an annealing process can be performed on the structure of FIG. **29** at a temperature of about 500° C. to about 800° C. in a hydrogen ambient.

(53) In some embodiments, the etching of the portions of oxide layer **2828** on dielectric layer **318** is not performed, and as a result, the structure of FIG. **29** is not formed and operations **220-240** are performed on the structure of FIG. **28**. Performing operations **220-240** on the structure of FIG. **28** can result in region **102A1** having oxide layer **2828** as the barrier layer, as shown in FIG. **34**, instead of barrier layer **128**, as shown in FIG. **1B**. In some embodiments, dielectric layer **818** that is formed in operation **220** can be formed with a front side surface **818f** substantially coplanar with a front side surface **2828f** of dielectric layer **2828**, as shown in FIG. **34**, or can be formed on front side surface **2828f**, as shown in FIG. **35**.

(54) In some embodiments, in operation **215** of FIG. **2**, instead of the oxidation process described to form barrier layers **128** with reference to FIG. **10**, barrier layers **128** can be formed as described with reference to FIGS. **30-33**. The formation of barrier layers **128** can include the sequential operations of (i) forming a patterned masking layer **3060** with openings **3062** on the structure of FIG. **9** to form the structure of FIG. **30**, (ii) depositing an oxide layer **3128** on the structure of FIG. **30** to form the structure of FIG. **31**, (iii) performing a lift-off process to remove patterned masking layer **3060** and portions of oxide layer **3128** on patterned masking layer **3060** to form oxide layer **3228** of FIG. **32**, and (iv) etching portions of oxide layer **3228** on sidewalls of dielectric layer **318** to form the structure of FIG. **33** on which operations **220-240** can be performed. In some embodiments, oxide layer **3128** can include an insulating material and can be deposited with a CVD process, an ALD process, or a suitable deposition process for depositing dielectric materials. In some embodiments, the etching of oxide layer **3228** can be performed using a wet etch process (e.g., using DHF) or a dry etch process. Following the formation of barrier layers **128**, an annealing process can be performed on the structure of FIG. **33** at a temperature of about 500° C. to about 800° C. in a hydrogen ambient.

(55) In some embodiments, the etching of the portions of oxide layer **3228** on sidewalls of dielectric layer **318** is not performed, and as a result, the structure of FIG. **33** is not formed and operations **220-240** are performed on the structure of FIG. **32**. Performing operations **220-240** on the structure of FIG. **32** can result in region **102A1** having oxide layer **3228** as the barrier layer, as shown in FIG. **36**, instead of barrier layer **128**, as shown in FIG. **1B**. In some embodiments, dielectric layer **818** that is formed in operation **220** can be formed with a front side surface **818f** substantially coplanar with a front side surface **3228f** of dielectric layer **3228** and a front side surface **318f** of dielectric layer **318**, as shown in FIG. **36**, or can be formed on front side surfaces **3228f** and **318f**, as shown in FIG. **37**.

(56) The present disclosure provides example methods of fabricating BSI image sensors (e.g., BSI image sensor **102**) with improved surface uniformity of Ge or SiGe epitaxial structures (e.g., epitaxial structure **124**) in pixel structures (e.g. pixel structure **122A-122**). In some embodiments, the example methods (e.g., method **200**) form barrier layers (e.g., barrier layer **128**) at the edges of the trenches (e.g., trench edge **108e**) in which the Ge or SiGe epitaxial structures are formed. The barrier layers can prevent the formation of Ge or SiGe layers at the trench edges because the barrier layers include dielectric materials, such as silicon oxide (SiO₂), that are not favorable for the growth of Ge or SiGe layers. The materials of the barrier layers can inhibit the growth of Ge or

SiGe layers on the barrier layers, and as a result, inhibits the growth of Ge or SiGe layers at the trench edges. Preventing the lateral expansion of the materials of the Ge or SiGe epitaxial structures can prevent the formation of recesses on the surfaces of the Ge or SiGe epitaxial structures.

(57) Thus, the barrier layers at the trench edges can minimize or eliminate surface non-uniformity, and consequently reduce or eliminate the formation of non-uniform interfaces between the Ge or SiGe epitaxial structures and silicon capping layers (e.g., capping layer **126**). In some embodiments, the surface non-uniformity in the Ge or SiGe epitaxial structures is reduced by about 50% to about 100% compared to epitaxial structures in BSI image sensors without the barrier layers. As a result, the sensor performance of the BSI image sensors with the barrier layers is increased by about 40% to about 60% compared to BSI image sensors without the barrier layers at the trench edges.

(58) In some embodiments, a method includes depositing a dielectric layer on a substrate, forming a trench within the dielectric layer and the substrate, forming an epitaxial structure within the trench, and forming a barrier layer with first and second layer portions. The first layer portion is formed on a sidewall portion of the trench that is not covered by the epitaxial structure. The method further includes forming a capping layer on the epitaxial structure and adjacent to the barrier layer, selectively doping regions of the epitaxial structure and the capping layer, selectively forming a silicide layer on the doped regions, depositing an etch stop layer on the silicide layer, and forming conductive plugs on the silicide layer through the etch stop layer.

(59) In some embodiments, a method includes depositing a dielectric layer on a substrate, forming an epitaxial structure within the substrate, depositing a semiconductor layer on the dielectric layer, the epitaxial structure, and a substrate region between the dielectric layer and the epitaxial structure, etching portions of the semiconductor layer from the dielectric layer and the epitaxial structure to form a barrier layer including a portion of the semiconductor layer on the substrate region, forming a capping layer on the epitaxial structure and adjacent to the barrier layer, selectively doping regions of the epitaxial structure and the capping layer, and forming contact structures on the doped regions.

(60) In some embodiments, a semiconductor device includes a substrate, a dielectric layer disposed on the substrate, and a pixel region with a pixel structure. The pixel structure includes an epitaxial structure disposed within the substrate, a capping layer disposed on the epitaxial structure, and a barrier layer disposed on a substrate region of the substrate adjacent to and surrounding the epitaxial structure. The barrier layer includes an oxide of a material of the substrate. The semiconductor device further includes an isolation region with an isolation structure disposed adjacent to the pixel region and a contact pad region with a pad structure disposed adjacent to the isolation region.

(61) The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device, comprising: a substrate comprising a top sidewall portion and a bottom sidewall portion; a dielectric layer disposed on the substrate; a pixel region comprising a pixel structure, wherein the pixel structure comprises: an epitaxial structure disposed in the substrate,

wherein a sidewall of the epitaxial structure is in physical contact with the bottom sidewall portion of the substrate; a capping layer disposed on a top surface of the epitaxial structure; and a barrier layer surrounding the capping layer and disposed on and in physical contact with a substrate region of the substrate that extends above the top surface of the epitaxial structure, wherein the barrier layer comprises an oxide of a material of the substrate region; an isolation region comprising an isolation structure disposed adjacent to the pixel region; and a contact pad region comprising a pad structure disposed adjacent to the isolation region.

2. The semiconductor device of claim 1, wherein a sidewall of the capping layer is spaced apart from a sidewall of the barrier layer by a portion of the dielectric layer.

3. The semiconductor device of claim 1, wherein a first sidewall portion of the capping layer is in contact with the barrier layer and a second sidewall portion of the capping layer is in contact with the dielectric layer.

4. The semiconductor device of claim 1, wherein the substrate region comprises the top sidewall portion of the substrate in physical contact with the barrier layer.

5. The semiconductor device of claim 1, wherein the substrate region comprises a surface portion of the substrate in contact with the dielectric layer.

6. The semiconductor device of claim 1, wherein the pixel structure further comprises doped regions disposed in the capping layer and the epitaxial structure.

7. The semiconductor device of claim 1, wherein a first portion of the barrier layer extends along a sidewall of the substrate region and a second portion of the barrier layer extends along a front side surface of the substrate region.

8. The semiconductor device of claim 1, wherein the dielectric layer is in contact with a front side surface of the capping layer.

9. The semiconductor device of claim 1, wherein the dielectric layer is in contact with a front side surface of the barrier layer.

10. A semiconductor device, comprising: a substrate; a pixel structure, comprising: a germanium-based epitaxial region disposed in the substrate, wherein sidewalls of the germanium-based epitaxial region is in physical contact with sidewalls of the substrate and wherein a top surface of the germanium-based epitaxial region is below a top surface of the substrate; a capping layer disposed on the top surface of the germanium-based epitaxial region; and an oxide layer surrounding the capping layer; and a doped isolation structure disposed adjacent to the pixel structure.

11. The semiconductor device of claim 10, wherein the germanium-based epitaxial region comprises a silicon germanium region or a germanium region.

12. The semiconductor device of claim 10, wherein the capping layer comprises a silicon layer.

13. The semiconductor device of claim 10, wherein the oxide layer is disposed between a sidewall of the capping layer and a sidewall of the substrate.

14. The semiconductor device of claim 10, wherein the oxide layer extends from a top surface of the substrate to a sidewall surface of the substrate.

15. The semiconductor device of claim 10, wherein a top surface of the oxide layer extends above top surfaces of the substrate and the capping layer.

16. The semiconductor device of claim 10, wherein the pixel structure further comprises: a p-type doped region disposed in the capping layer; and an n-type doped region disposed in the capping layer.

17. A semiconductor device, comprising: a substrate; a dielectric layer disposed on the substrate; an epitaxial structure disposed in the substrate; a barrier layer disposed on a substrate region between the dielectric layer and a top surface of the epitaxial structure, wherein the substrate region extends above a top surface of the epitaxial structure; and a capping layer disposed on the top surface of the epitaxial structure, wherein a sidewall of the capping layer is spaced apart from a sidewall of the barrier layer by a portion of the dielectric layer.

18. The semiconductor device of claim 17, wherein the capping layer comprises a silicon layer.
 19. The semiconductor device of claim 17, wherein the epitaxial structure comprises a silicon germanium region or an undoped germanium region.
 20. The semiconductor device of claim 17, wherein the barrier layer surrounds the capping layer.
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